Find derivative of $f(z) = (iz^3 + 3z^2)^3$ at $z = 2i$.

~~Let $iz^3 + 3z^2 = w$~~

~~$f(z) = w^3$~~

~~$\frac{d}{dz} f(z)$~~

$$f(z) = (iz^3 + 3z^2)^3$$

$$f'(z) = \frac{d}{dz}(iz^3 + 3z^2)^3$$

$$= 3(iz^3 + 3z^2)^2 (3z^2 i + 6z)$$

at $z = 2i$

$$= 3[i(2i)^3 + 3(2i)^3]^2 (3(2i)^2 i + 6(2i)$$

$$= 3[i(8i^3) + 3(4i^2)]^2 (12i^3 + 12i)$$

$$= 3(8i^4 + 12i^2)^2 (12i^3 + 12i)$$

$$= 3(+8 - 12)^2 (-12i + 12i)$$

$$= 3(16)(0)$$

$$= 0.$$

$$\lim_{z \to 4i} \frac{z^2+16}{z-4i}$$

$$\lim_{z \to 4i} \frac{z^2-(-16)}{z-4i}$$

$$\lim_{z \to 4i} \frac{z^2-16i}{z-4i}$$

$$\lim_{z \to 4i} \frac{(z)^2-(4i)^2}{z-4i}$$

$$\lim_{z \to 4i} \frac{(z+4i)(z-4i)}{z-4i}$$

$$= \lim_{z \to 4i} z+4i$$

$$= 4i+4i$$

$$= 8i$$

Evaluate $(-4-4i)^3$

$z = -4-4i$

$r = \sqrt{(4)^2+(4)^2}$

$= \sqrt{16+16} = \sqrt{32}$

$= 4\sqrt{2}$

$\theta = \tan^{-1}(y/x)$

$= \tan^{-1}(1)$

$= \pi/4$

both are -ive

$Arg(z) = \theta = \pi + \theta$

$= \pi + \pi/4 = 5\pi/4$

$z = -4-4i = (4\sqrt{2}e^{i5\pi/4})$

$(-4-4i)^3 = (4\sqrt{2}e^{i5\pi/4})^3$.

$$\lim_{z \to 2+i} \; 2z^2 + 4 \cdot 3z + 5i$$

$$= 2(2+i)^2 + 4 - 3(2+i) + 5i$$
$$= 2(4 + 4i + i^2) + 4 - 6 - 3i + 5i$$
$$= 2(4 + 4i - 1) + 4 - 6 - 3i + 5i$$
$$= 8 + 8i - 2 + 4 - 6 - 3i + 5i$$
$$= 8 - 8 + 10i - 3i + 4$$
$$= 6i + 4.$$

or $$= 4 + 10i$$

$$\lim_{z \to 2+i} (2z^2 + 4 - 3z + 5i).$$

$$= 2(2+i)^2 + 4 - 3(2+i) + 5i$$
$$= 2(4 + 4i - 1) + 4 - 6 - 3i + 5i$$
$$= 8 + 8i - 2 + 4 - 6 - 3i + 5i$$
$$= 10i + 4$$

or $$= 4 + 10i.$$

Find $(1+\sqrt{3}i)^{\frac{1}{3}}$

exp-form was given.

$$1+\sqrt{3}i = r\,2e^{i\pi/3}$$

$$(1+\sqrt{3}i)^{\frac{1}{3}} = (2e^{i\pi/3})^{\frac{1}{3}}$$

$$= \sqrt[3]{2}\, e^{i\pi/9}$$

$$= \sqrt[3]{2}\left(\cos\frac{\pi}{9} + i\sin\frac{\pi}{9}\right)$$

$$= \sqrt[3]{2}(0.9397 + i(0.3240)$$

Express in exponential form.

$z = \sqrt{3} - i$

$r = \sqrt{(\sqrt{3})^2 + (-1)^2}$

$r = \sqrt{3+1} = \sqrt{4} = 2$

$\boxed{r = 2}$

$\theta = \tan^{-1}(y/x)$ ignoring -ve sign.

$= \tan^{-1}(1/\sqrt{3})$

$= \pi/6$

As $x$ is +ive & $y$ is -ve

$\theta$ Arg(z) lies in 4th quad.

so $\theta = -\theta$

$\theta = -\frac{\pi}{6}$

$z = re^{i\theta}$

$= 2e^{-i\pi/6}$.

$$f(z) = \frac{z-1}{z+1} \quad \text{at} \quad z = i$$

$$f'(z) = \frac{d}{d(z)}\left(\frac{z-1}{z+1}\right)$$

$$= \frac{(z+1)(1) - (z-1)(1)}{(z+1)^2}$$

$$= \frac{z+1-z+1}{(z+1)^2}$$

$$= \frac{2}{(z+1)^2}$$

$$\lim_{z \to i} \frac{2}{(z+1)^2} \Rightarrow \frac{2}{(i+1)^2}$$

$$= \frac{2}{i^2+2i+1} = \frac{2}{-1+2i+1}$$

$$= \frac{2}{2i} = \frac{1}{i}$$ Ans

Find the value of
$(1+\sqrt{3}i)^4$

$1+\sqrt{3}i$

$r = \sqrt{1+3} = 2$

$\theta = \tan^{-1}\sqrt{3}$

$\theta = \pi/3$

x, y both are +ive

So $\alpha\ \theta = \theta$

$\theta = \pi/3$

$1+\sqrt{3}i = 2re^{i\pi/3}$

$(1+\sqrt{3}i)^4 = (2e^{i\pi/3})^4$

$= 16e^{i4\pi/3}$

$= 16\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$

$= 16\left(\frac{-1}{2} - i\frac{\sqrt{3}}{2}\right)$

$= 8(-1-\sqrt{3}i)$

$= -8(1+\sqrt{3}i)$.

$= \sqrt[3]{2}\, e^{i\pi/9}$

$= \sqrt[3]{2}\left(\cos\frac{\pi}{9} + i\sin\frac{\pi}{9}\right)$

$= \sqrt[3]{2}(0.9397 + i(0.3240)$

$f(y) = (y+6y^2)^3$

$g(y) = y+6y^2$

$h(y) =$

$f(y) = [g(y)]^3$ $\qquad [g(y)]^3 = f(y)$

$\frac{d}{dy}[g(y)]^3 = 3\frac{d}{dy}[g(y)]^2\, g'(y)$

$= 3\frac{d}{dy}\, y+6y^2$

Evaluate:-

$$\lim_{z \to \infty} \frac{z-1}{z^2+2+1} \quad \left(\frac{\infty}{\infty}\right)$$

Applying L'Hospital Rule.

$$= \lim_{z \to \infty} \frac{\frac{d}{dz}(z-1)}{\frac{d}{dz}(z^2+2+1}$$

$$= \lim_{z \to \infty} \frac{1-0}{2z+1}$$

$$= \lim_{z \to \infty} \frac{1}{2z+1}$$

$$= \frac{1}{\infty} = 0.$$ Converges

$$\lim_{z \to i} \frac{z-1}{z^2+1}$$

$f'(z) = ?$   $ru_r = v_\theta$   $u_\theta = -rv_r$

$$f(z) = \sqrt{r}\, e^{i\theta/2}$$

$$= \sqrt{r}\cos\frac{\theta}{2} + i\sqrt{r}\sin\frac{\theta}{2}$$

so

$$u(r,\theta) = \sqrt{r}\cos\frac{\theta}{2}$$

and

$$v(r,\theta) = \sqrt{r}\sin\theta/2$$

$$v_\theta = \frac{\sqrt{r}}{2}\cos\frac{\theta}{2}$$

$$v_r = \frac{1^2}{2\sqrt{r}}\sin\frac{\theta}{2}$$

CR Eqs in Polar Form are

① $ru_r = v_\theta$

② $u_\theta = -rv_r$

① $u_r = \frac{1}{2\sqrt{r}}\cos\frac{\theta}{2}$

$$ru_r = \frac{r}{2\sqrt{r}}\cos\frac{\theta}{2} = \frac{\sqrt{r}}{2}\cos\frac{\theta}{2} = v_\theta \checkmark$$

② $u_\theta = \frac{-1}{2}\sqrt{r}\sin\frac{\theta}{2}$

$$= -rv_r \checkmark$$

$\therefore f'(z) = e^{-i\theta}(u_r + iv_r)$.

$$f'(z) = e^{-i\theta}\left(\frac{1}{2\sqrt{r}}\cos\frac{\theta}{2} + i\frac{1}{2\sqrt{r}}\sin\frac{\theta}{2}\right).$$

$$= \frac{1}{2\sqrt{r}}e^{-i\theta}(\cos\theta/2 + i\sin\theta/2)$$

$$= \frac{1}{2\sqrt{r}}e^{-i\theta}e^{\theta/2} \Rightarrow \frac{1}{2\sqrt{r}}\cdot e^{-i\theta/2} = \frac{1}{2\sqrt{r}e^{i\theta/2}}$$

$$f'(z) = \frac{1}{2f(z)}$$

$(1+z)^2 = 1+z^2+2z$

L.H.S $= (1+z)^2$

$= (1+z)(1+z)$

$= 1(1+z)+z(1+z)$

$= 1+z+z+z^2$

$= 1+2z+z^2$

$= 1+z^2+2z$ . = R.H.S

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luate $(-4-4i)^3$

$-4-4i$

$\sqrt{(4)^2+(4)^2}$

$\sqrt{16+16} = \sqrt{32}$

$\theta = \tan^{-1}(y$

$= \tan^{-1}(1$

C-R equations

$f(z) = z - \bar{z}$

$= x + iy - (x - iy)$

$= x + iy - x + iy$

$f(z) = 2iy$

$u = 0 \quad , \quad v = 2y$

$u_x = 0 \quad , \quad v_x = 0$

$u_y = 0 \qquad v_y = 2$

$u_x = v_y \quad , \quad u_y = -v_x$

$0 \neq 2 \qquad 0 = 0$

as C-R eqs are not satisfied.

so $f'(z)$ does not exist at any point.

$f(y) = (y+6y^2)^3$

$f'(y) = ?$

$$f'(y) = \frac{d}{dy}(y+6y^2)^3$$

$$= 3(y+6y^2)^{3-1} \frac{d}{dy}(y+6y^2)$$

$$= 3(y+6y^2)^2 \left[\frac{d}{dy}(y) + 6\frac{d}{dy}y^2\right]$$

$$= 3(y+6y^2)^2 (1 + 6(2y))$$

$$= 3(y+6y^2)^2 (1+12y).$$

(1) $\lim_{z \to z_0} f(z)$ exist

$z = x + iy$ , $y = mx$

$$\Rightarrow \frac{z^3}{2\operatorname{Re} z} = \frac{z^2}{x} = \frac{x^2 - y^2 + 2ixy}{x}$$

$$= \frac{x^2 - m^2x^2 + 2imx^2}{x}$$

$$= x - m^2x + 2imx.$$

$$= x(1 - m^2 + 2im).$$

$$= \lim_{z \to 0} x(1 - m^2 + 2im).$$

$$= 0(1 - m^2 + 2im) = 0.$$

(2) $f(z_0)$ exist

$f(0) = 0$

(3) $\lim_{z \to z_0} f(z_0) = f(z_0)$.

all conditions are satisfied. so $f$ is continuous.

Reduce to Real form.

$$\frac{1+2i}{3-4i} \cdot \frac{2-i}{5i}$$

$$= \frac{5i(1+2i)+(2-i)(3-4i)}{5i(3-4i)}$$

$$= \frac{5i+10+(6-4-8i-3i)}{15i-20i^2}$$

$$= \frac{5i-10+2-11i}{15i+20}$$

$$= \frac{-6i-8}{15i+20} \cdot \frac{-15i+20}{-15i+20}$$

$$= \frac{(-6i-8)(-15i+20)}{+400+225}$$

$$= \frac{+90i^2-120i+120i-160}{625} \Rightarrow \frac{-250}{625} = -\frac{2}{5}$$

$$= \frac{-90-160}{625}$$

$$= \frac{-250}{625}$$

$$= -\frac{2}{5}$$

$$(20+15i)(20-15i)$$

$$(20)^2-(15i)^2$$

$z = x + iy$

$x = \frac{z+\bar{z}}{2}$ ; $y = \frac{z-\bar{z}}{2i}$.

$z = x + iy$

$\bar{z} = x - iy$

$z + \bar{z} = x + iy + x - iy$

$z + \bar{z} = 2x$

$\frac{z+\bar{z}}{2} = x$

$z = x + iy$

$z = x - iy$

$z - \bar{z} = x + iy - x + iy$

$z - \bar{z} = 2iy$

$\frac{z-\bar{z}}{2i} = y$.

$$f(z) = \frac{z^3+3z+2}{z^3+1} \quad \text{at } z=i$$

If $f$ and $g$ are continuous at $z_0$ then $f/g$ is cont at $z_0$.

$g(z_0) \neq 0$

But here

$$g(z_0) = i^3+1$$
$$= -i+1 = 0$$

so $f(z)$ is discontinuous.

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$$f(z) = \frac{1}{z} \quad \text{at } z = -2i$$

$$\lim_{z \to z_0} f(z) = \lim_{z \to -2i} \frac{1}{z}$$

$$= \frac{1}{-2i} \cdot \frac{i}{i} = \frac{i}{-2i^2} = \frac{i}{2}$$

$$f(z_0) = \frac{1}{-2i}$$

$$\lim_{z \to z_0} f(z) = f(z_0) \neq \frac{i}{2}.$$

Evaluate.

$$= \lim_{z \to i} \frac{z-i}{z^2-(-1)}$$

$$= \lim_{z \to i} \frac{z-i}{z^2-i^2}$$

$$= \lim_{z \to i} \frac{z-i}{(z+i)(z-i)}$$

$$= \lim_{z \to i} \frac{1}{z+i}$$

$$= \frac{1}{i+i}$$

$$= \frac{1}{2i}$$

Express in exponential form.

$z = -1 + i$

$x = -1$ and $y = 1$

$|z| = r = \sqrt{x^2 + y^2}$

$= \sqrt{(-1)^2 + (1)^2}$

$r = \sqrt{1+1} = \sqrt{2}$

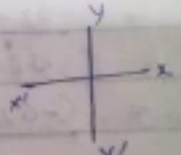
$\theta = \tan^{-1}(y/x)$

$= \tan^{-1}(1/1)$

$\theta = \tan^{-1}(1) = \pi/4$

As $x$ is -ive & $y$ is +ive

so, Arg(z) lies in 2nd quadrant, where $\theta = \pi - \theta$




$\theta = \pi - \pi/4$

$= \frac{4\pi - \pi}{4}$

$= \frac{3\pi}{4}$.

$z = re^{i\theta}$

$z = \sqrt{2}e^{i3\pi/4}$